

# torch.geqrf#

<https://pytorch.org/docs/stable/generated/torch.geqrf.html>

## Description:

This is a low-level function for calling LAPACK's `geqrf` directly. This function returns a namedtuple `(a, tau)` as defined in LAPACK documentation for `geqrf`. Computes a QR decomposition of input. Both Q and R matrices are stored in the same output tensor `a`. The elements of R are stored on and above the diagonal. Elementary reflectors (or Householder vectors) implicitly defining matrix Q are stored below the diagonal. The results of this function can be used together with `torch.linalg.householder_product()` to obtain the Q matrix or with `torch.ormqr()`, which uses an implicit representation of the Q matrix, for an efficient matrix-matrix multiplication. See LAPACK documentation for `geqrf` for further details. See also `torch.linalg.qr()`, which computes Q and R matrices, and `torch.linalg.lstsq()` with the `driver="gels"` option for a function that can solve matrix equations using a QR decomposition.

## Function Signature

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```
torch.geqrf(input, *, out=None)#
```

Parameters

Keyword Arguments

## Parameters

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### Keyword

out (tuple, optional) – the output tuple of (Tensor, Tensor). Ignored if None.  
Default: None.

## Notes & Warnings

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**NOTE** See also `torch.linalg.qr()`, which computes Q and R matrices, and `torch.linalg.lstsq()` with the `driver="gels"` option for a function that can solve matrix equations using a QR decomposition.

# Detailed Documentation

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## Documentation

This is a low-level function for calling LAPACK's `geqrf` directly. This function returns a `namedtuple` (`a`, `tau`) as defined in LAPACK documentation for `geqrf`.

Computes a QR decomposition of input. Both Q and R matrices are stored in the same output tensor `a`. The elements of R are stored on and above the diagonal. Elementary reflectors (or Householder vectors) implicitly defining matrix Q are stored below the diagonal. The results of this function can be used together with `torch.linalg.householder_product()` to obtain the Q matrix or with `torch.ormqr()`, which uses an implicit representation of the Q matrix, for an efficient matrix-matrix multiplication.

See LAPACK documentation for `geqrf` for further details.

See also `torch.linalg.qr()`, which computes Q and R matrices, and `torch.linalg.lstsq()` with the `driver="gels"` option for a function that can solve matrix equations using a QR decomposition.

`input` (Tensor) – the input matrix

`out` (tuple, optional) – the output tuple of (Tensor, Tensor). Ignored if None. Default: None.